Project Report: Quiz Master (Modern Application Development - 1)

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1. Project Statement:

Title: Quiz Master

Description: The Quiz Master is a quiz management web-based app. The goal of this project is to create an interactive platform where admins can create, manage, and track quizzes while users (students) can attempt quizzes, view results, and track their performance.

Objectives:

- Develop an interactive quiz platform for students.
- Enable administrators to manage subjects, quizzes, and users.
- Implement authentication and role management.
- Provide summary charts and data visualization.
- Ensure security, UI responsiveness, and smooth API access.

2. Frameworks:

- Flask for application code
- Jinja2 templates for HTML generation and styling
- Flask-SQLAlchemy for data management
- Matplotlib for plotting graphs

3. Process, Challenges and Solutions:

- Started with making simple html pages as per the design given in the project wireframe and slowly added CSS and bootstrap to make things look good
- Added routes for the functionalities and linked them with respective html pages
- Faced problems in making the relationships of models, solved it by learning it from the sqlalchemy official website
- Initially had problems in creating summary charts refered some youtube videos for that

4. Functionalities:

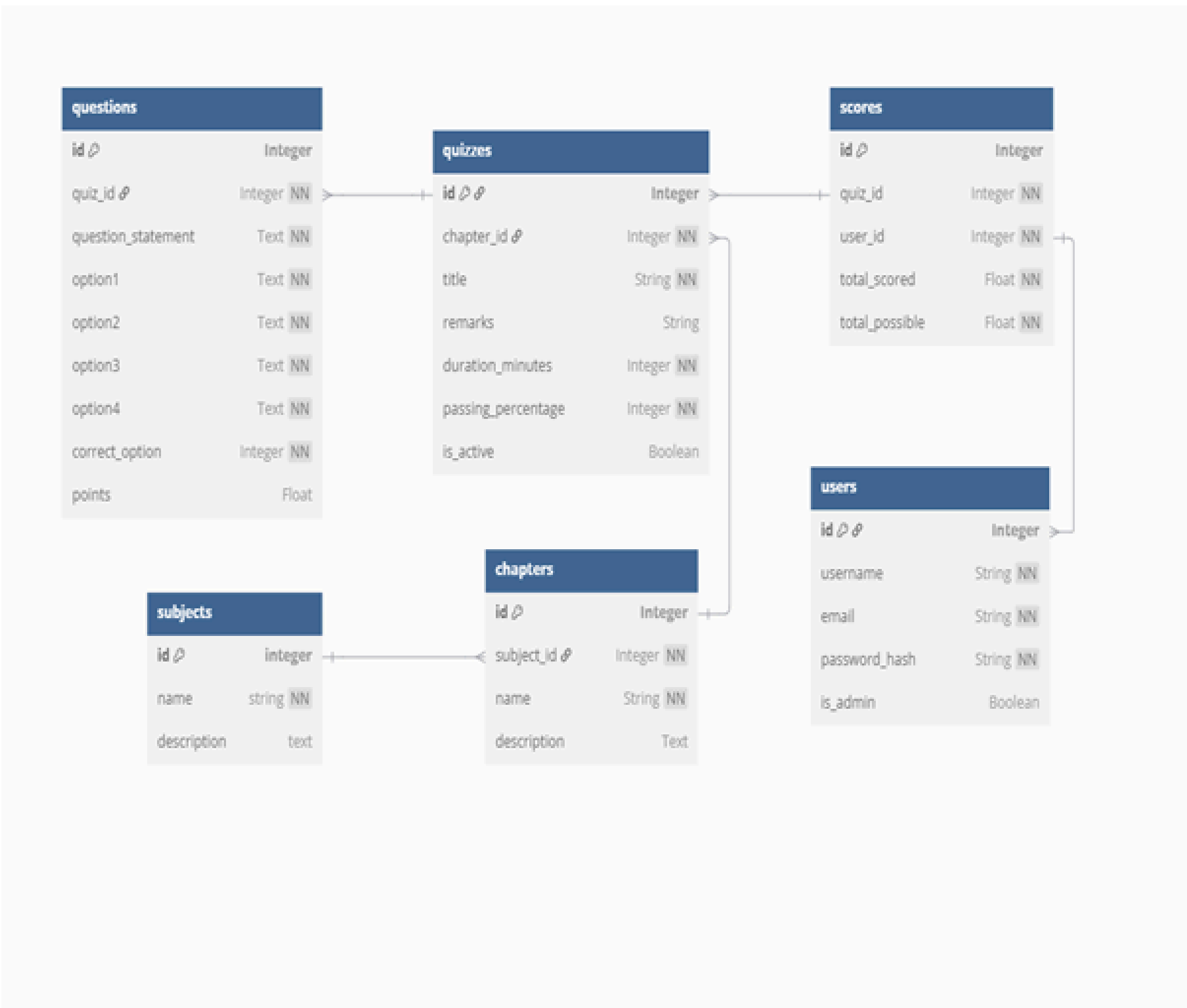
Admin Functionalities

- User Management: View a list of registered users.
- Quiz Management: Create, edit, and delete subjects, chapters, quizzes, and questions.
- Quiz Scheduling: Set time duration and availability for quizzes.
- Search: Find users, subjects, quizzes, and questions easily.
- Performance Analytics: View summary charts of quiz statistics and user performance.

User (Student) Functionalities

- Quiz Participation: Browse available quizzes and attempt them.
- Timer Functionality: Complete quizzes within a set time limit.
- Instant Feedback: View correct/incorrect answers after submission.
- Score Tracking: Check past quiz attempts and performance history.
- Search: Look up subjects and quizzes easily.

5. E-R Diagram:



Database Structure Summary

Your application appears to use a relational database with several interconnected tables:

1. **Users**
    - Fields: id, username, email, password_hash, is_admin
    - Stores both regular users and administrators
    - Authentication is done by comparing password_hash directly
2. **Subjects**
    - Fields: id, name, description
    - Managed exclusively by administrators
3. **Chapters**
    - Fields: id, subject_id, name, description
    - Organized under subjects
    - Each chapter belongs to exactly one subject
4. **Quizzes**
    - Fields: id, chapter_id, title, remarks, duration_minutes, passing_percentage, is_active
    - Each quiz belongs to a chapter
5. **Questions**
    - Fields: id, quiz_id, question_statement, option1, option2, option3, option4, correct_option, points
    - Multiple-choice questions with 4 options
    - Each question belongs to one quiz
    - Questions can have different point values
6. **Scores**
    - Fields: id, quiz_id, user_id, total_scored, total_possible
    - Records user performance on quizzes
    - Links users to quizzes they've completed
    - Stores both the achieved score and maximum possible score

The database follows a hierarchical structure:

- Subjects contain Chapters
- Chapters contain Quizzes
- Quizzes contain Questions
- Users take Quizzes, producing Scores

6. Presentation link:

Quiz_master presentation.mkv

https://drive.google.com/file/d/1e2rctYq1NlPuD53fqf1Fki8hUws2PwhW/view?usp=sharing